

MODULE 1 – QUANTUM CHEMISTRY

Quantum chemistry is a branch of Chemistry that deals with the applications of principles of quantum mechanics to the study of the properties of molecules. The central topic in quantum chemistry is Understanding electronic structure and physical movement of atoms and molecules using the Schrödinger equations.

Mechanics is the branch of science which deals with the study of the motion of objects. Quantum mechanics deals with the laws of motion and the behaviour of particles at the microscopic level. But **classical mechanics** explains the properties of bulk matter using the principles of thermodynamics, Newton's laws of motion, etc. It explains the bonding between atom by considering the electrons as rigid particles whose position and momentum can be exactly determined at any point of time.

Later it was realized that the laws of classical mechanics were not in agreement with the observed features of motion of microscopic particles like electrons. Apart from this, black body radiation spectrum, the photo electric effect and emission of spectral lines from atoms could not be explained within the framework of classical mechanics. These failures of classical mechanics led to the development of a new concept called quantum mechanics which was successful in explaining the behaviour of atoms, molecules, nuclei and electrons. In classical mechanics the dynamical variables like position, momentum, etc, can be determined or predicted with certainty. But in quantum mechanics the dynamical variables cannot be determined or predicted with certainty. For example, according to Bohr model of atom, the radius of the electron's orbit in a ground state hydrogen atom is exactly $5.3 \times 10^{-11} \text{m}$. But according to quantum mechanics that is the most probable radius.

The fundamental concepts on which quantum mechanics is based are the de Broglie's postulates, Heisenberg's uncertainty principle and the Schrodinger's wave equations with their solutions.

Matter waves

Newton thought that light was made up of particles, but then it was discovered that it behaves like a wave in the case of interference and diffraction phenomenon. Later however in the beginning of the twentieth century it was found that light indeed behave like a particle in the case of photoelectric effect.

In 1924-25 Louis De- Broglie proposed that if radiation exhibits dual nature, i.e., properties of both waves and particles, then matter which consists of particles must also possess wave properties. This is called De-Broglie hypothesis. The waves associated with particles of matter are called matter waves.

De-Broglie wave equation

According to Max Planck's quantum theory of radiation, the energy of a photon is

$$E = h\nu \quad \rightarrow (1)$$

Where h is called Planck's constant and ν is the frequency.

But according to Einstein's mass-energy relationship, energy associated with a particle is

$$E = mc^2 = pc \quad \rightarrow (2)$$

where p is the momentum. Equating equations (1) and (2) we get

$$pc = h\nu \quad \text{or} \quad p = \frac{h\nu}{c}$$

$$\text{But } (\nu = \frac{c}{\lambda})$$

$$\text{i.e., } p = \frac{h}{\lambda}$$

or $\lambda = \frac{h}{p}$ or $\lambda = \frac{h}{mv}$ where m is the mass of the particle and v its velocity.

This is the famous De- Broglie wave equation connecting particle property (mv) and wave property (λ)

Following De-Broglie's postulate, in 1927 **Heisenberg** gave the **uncertainty principle** which stated that it is impossible to determine simultaneously the position and momentum (or velocity) of a small moving particle like electron at any given instant with absolute certainty.

Mathematically, the uncertainty principle may be expressed as

$\Delta x \times \Delta p \geq \frac{h}{4\pi}$ where Δx and Δp are respectively the uncertainties in position and momentum of a moving particle while h is the Planck's constant (6.626×10^{-34} Joule sec.). Thus, according to Heisenberg, since one cannot know the precise position and momentum of a particle at a given instant, one cannot calculate the precise future motion of a particle, but only a range of possibilities for the future motion of the particle. Since experimental evidence suggested that atomic particles were also wave-like in nature, Erwin Schrodinger in 1926 arrived at a wave equation to explain the behaviour of atomic particles. The Schrodinger's wave equation is a second derivative differential equation given as follows:

$\frac{\partial^2 \psi}{\partial x^2} + \frac{8\pi^2 m}{h^2} (E-V)\psi = 0$ where $\frac{\partial^2 \psi}{\partial x^2}$ is the second derivative with respect to x, ψ is the Schrodinger wave function, x is the position, E is the total energy, and V is the potential energy.

Nature and significance of wave function, ψ

Wave function is a mathematical representation of a wave associated with particles of matter. It describes state of a particle as a mathematical function of position and time. i.e., $\psi = f(x,t)$ where x is the position coordinate and t, time. The physical interpretation of ψ is that it is a measure of the amplitude of the matter wave associated with the particle at various positions and time intervals. It also represents a measure of probability of finding a particle at a particular position at a particular instant. In general the wave function ψ is a complex quantity whereas the probability is real and positive. Therefore a term called probability density defined as the square of the absolute value of ψ , ($|\psi|^2$), is used to represent the probability of finding the particle at a particular position and at a particular time. Thus wave function can also be described as the square root of the probability of finding a particle within a certain domain of space.

Operators in quantum mechanics

In mathematics an operator is one which operates on a function and transforms it into another function. A familiar example is the differential operator d/dx which when applied to a function f(x), we get the differential coefficient of f(x), denoted by f'x.

e.g., $\frac{d}{dx} 4x^3 = 12x^2$

In quantum mechanics each dynamical variable (measurable physical quantities such as position, momentum, total energy, etc.) is represented by an operator. In classical physics, the dynamical variables can be measured with certainty. But in quantum mechanics, there is uncertainty in such measurements. Such measurements can be accomplished only by carrying out some operations like differentiation, etc., on the wave function Ψ . E.g., the operation to be carried out to find the position of a particle is the position operator, represented as $\hat{x}$. Other important operators are momentum operator $\hat{p}$, potential energy operator $\hat{V}$ and total energy operator or Hamiltonian operator $\hat{H}$.

Eigen values and eigen functions of operators

If an operator $\hat{A}$ operating on a function Ψ gives the same function multiplied by a constant factor λ , i.e., if $\hat{A} \Psi = \lambda \Psi$, then, Ψ is called the eigen (eigen means proper or characteristic) function of the operator $\hat{A}$ and the constant factor λ is called the eigen value. The equation

$\hat{A} \Psi = \lambda \Psi$ is called eigen value equation for the operator $\hat{A}$.

Derivation of the Schrodinger wave equation-

The Schrodinger equation, named after ERWIN SCHRODINGER, is the most fundamental equation of Quantum mechanics.

Let us consider the vibration of a stretched string. The mathematical function which represents a wave is called wave function and is denoted by Ψ (psi). The equation for the wave function of such a simple harmonic wave on a string is represented as,

$$\Psi = A \sin \frac{2\pi x}{\lambda} \quad \text{-----eq 1}$$

Ψ = wave function ; x = displacement ; λ = wave length ; A = amplitude of the wave

Differentiating the eq (1) with respect to ' x ' ,

$$\frac{\partial \Psi}{\partial x} = A \left[\cos \frac{2\pi x}{\lambda} \right] \left(\frac{2\pi}{\lambda} \right)$$

$$\frac{\partial \Psi}{\partial x} = \left(\frac{2\pi A}{\lambda} \right) \left[\cos \frac{2\pi x}{\lambda} \right] \quad \text{----- eq 2}$$

Again differentiating the eq (2) with respect to ' x ' ,

$$\frac{\partial^2 \Psi}{\partial x^2} = \left(\frac{2\pi A}{\lambda} \right) \left[-\sin \frac{2\pi x}{\lambda} \right] \left(\frac{2\pi}{\lambda} \right)$$

$$\frac{\partial^2 \Psi}{\partial x^2} = \left(-\frac{4\pi^2 A}{\lambda^2} \right) \left[\sin \frac{2\pi x}{\lambda} \right] = \left(-\frac{4\pi^2}{\lambda^2} \right) \left[A \sin \frac{2\pi x}{\lambda} \right] \quad \text{----- eq 3}$$

putting the value of ' Ψ ' from eq (1) to eq (3)

$$\Psi = A \sin \frac{2\pi x}{\lambda}$$

$$\frac{\partial^2 \Psi}{\partial x^2} = \left(-\frac{4\pi^2 \Psi}{\lambda^2} \right) \quad \text{----- eq 4}$$

This equation is applicable to waves associated with all particles of like electrons , protons.

According to De Broglie equation,

$$\lambda = \frac{h}{mu}$$

$$\frac{1}{\lambda} = \frac{mu}{h}$$

$$\frac{1}{\lambda^2} = \frac{m^2 u^2}{h^2} \quad \text{--- eq 5}$$

putting the value of ' $\frac{1}{\lambda^2}$ ' from eq (5) to eq (4)

$$\frac{\partial^2 \psi}{\partial x^2} = \left(-\frac{4\pi^2 \psi m^2 u^2}{h^2} \right) \quad \text{---eq (6)}$$

$$\text{Kinetic energy} = \frac{mu^2}{2}$$

Total energy E = potential energy + Kinetic energy

$$E = V + \frac{mu^2}{2}$$

$$mu^2 = 2 (E - V) \quad \text{---eq 7}$$

putting the value of ' mu^2 ' from eq (7) to eq (6)

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{4\pi^2 m \cdot 2(E - V) \psi}{h^2}$$

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{8\pi^2 m (E - V) \psi}{h^2} \quad \text{--- eq 8}$$

This is the wave equation for the particle moving along the x- axis.

Eq (8) may be extended in three directions x , y , z. Hence,

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = -\frac{8\pi^2 m (E - V) \Psi}{h^2}$$

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} + \frac{8\pi^2 m (E - V) \Psi}{h^2} = 0 \quad \text{--- eq 9}$$

This is the time- independent Schrodinger equation for the wave associated with a particle moving in three dimensions.

Eq (9) may also be written as,

$$\nabla^2 \Psi + \frac{8\pi^2 m (E - V) \Psi}{h^2} = 0 \quad \text{--- eq 10}$$

This is also a form of Schrodinger wave equation.

∇^2 is Laplacian operator

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} = \nabla^2$$

$$-\nabla^2 \psi = \frac{8\pi^2 m (E-V) \psi}{h^2}$$

$$- \frac{\nabla^2 \psi h^2}{8\pi^2 m} = (E-V)\psi$$

$$- \frac{\nabla^2 \psi h^2}{8\pi^2 m} = E\psi - V\psi$$

$$- \frac{\nabla^2 \psi h^2}{8\pi^2 m} + V\psi = E\psi$$

$$\left[\left(- \frac{\nabla^2 h^2}{8\pi^2 m} \right) + V \right] \psi = E\psi \quad \text{---eq 11}$$

Schrodinger wave equation can also be expressed as

$$H\psi = E\psi \quad \text{---eq 11}$$

$$\text{Where } H = \left[V - \left(\frac{\nabla^2 h^2}{8\pi^2 m} \right) \right]$$

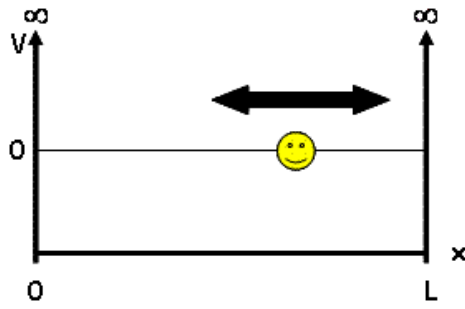
H = Hamiltonian operator , E = Eigen value

ψ (wave function) has no physical significance, ψ only represents the amplitude of the electron wave. ψ^2 represents the probability of locating an electron associated with a specific energy.

Application of Schrodinger equation to particle in a one dimensional box

The particle in a box problem is a common application of Schrodinger equation to a simplified system consisting of a particle moving horizontally within an infinitely deep well from which it cannot escape. Here our aim is to determine the wave function ψ and the allowed energy values of E that the particle can possess.

Consider a particle of mass m moving inside a one dimensional box of height ∞ and width L. Since the box has infinite height the particle cannot come outside rather it moves back and forth inside the box.



The potential energy is 0 inside the box ($V=0$ for $0 < x < L$) and goes to infinity (∞) at the walls of the box and outside the box ($V=\infty$ for $x < 0$ or $x > L$).

The Schrodinger equation for this single particle is

$$\frac{\partial^2 \Psi}{\partial x^2} = - \frac{8\pi^2 m (E-V) \Psi}{h^2}$$

Inside the box, $V=0$. So the above equation becomes

$$\frac{\partial^2 \Psi}{\partial x^2} = - \frac{8\pi^2 m E \Psi}{h^2} = - \frac{2m}{\hbar^2} E \Psi \quad \text{where } \hbar = \frac{h}{2\pi} \text{ (Reduced Planck's constant)}$$

If we put $\frac{2m}{\hbar^2} E = k^2$, then

$$\frac{\partial^2 \Psi}{\partial x^2} = - k^2 \Psi$$

This linear second-order differential equation can be solved in the same way that a algebraic quadratic equation is solved. It is separated into two factors, and each first-order differential equation can be integrated.

The general solution to this equation is

$\Psi = A \sin kx + B \cos kx \rightarrow (1)$ where A and B are constants to be evaluated from the boundary conditions. According to the boundary conditions, the probability of finding the particle at $x=0$ or $x=L$ is zero. i.e., Ψ must be zero at $x=0$ and $x=L$

Thus, applying the boundary condition that $\Psi = 0$ at $x=0$, to equation(1), we get

$$0 = A \sin k0 + B \cos k0 = B$$

Thus equation(1) becomes

$$\Psi = A \sin kx \rightarrow (2)$$

Applying the boundary condition that $\Psi = 0$ at $x = L$ to equation (2) we get

$$0 = A \sin kL$$

$$\text{Or } \sin kL = 0$$

But we know that

$$\sin n\pi = 0 \text{ for } n = 1, 2, 3, \dots$$

Comparing the two we get,

$$kL = n\pi$$

$$\text{or } k = \frac{n\pi}{L} \rightarrow (3)$$

Putting this in equation (2), we get

$$\Psi = A \sin \frac{n\pi}{L} x \rightarrow (4)$$

To determine A, remember that the total probability of finding the particle inside the box is 1. When we find the probability and set it equal to 1, the wave function is said to be normalized. We can now evaluate A by applying the normalization condition,

$$\text{i.e., } \int_0^L |\Psi|^2 dx = 1$$

$$\text{i.e., } \int_0^L A \sin \frac{n\pi}{L} x A \sin \frac{n\pi}{L} x dx = 1$$

$$\text{i.e., } A^2 \int_0^L \sin^2 \frac{n\pi}{L} x dx = 1$$

$$\text{But we know that } \sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\text{i.e., } \frac{A^2}{2} \int_0^L 1 - \cos \frac{2n\pi x}{L} dx = 1$$

$$\frac{A^2}{2} \left[x - \frac{\sin \frac{2n\pi x}{L}}{\frac{2n\pi}{L}} \right]_0^L = 1$$

$$\frac{A^2}{2} \left[L - \frac{\sin 2n\pi}{\frac{2n\pi}{L}} \right] = 1$$

$$\text{i.e., } \frac{A^2 L}{2} = 1$$

$$\text{Or } A^2 = \frac{2}{L}$$

$$\text{Or } A = \sqrt{\frac{2}{L}}$$

So the normalized wave functions of the particle are

$$\Psi = \sqrt{\frac{2}{L}} \sin \frac{n\pi}{L} x \quad \rightarrow (5)$$

The possible values of n are 1,2,3, etc. If n = 0, $\Psi = 0$ & $|\Psi|^2 = 0$, which is not possible.

Allowed energy levels (Eigen values) of the particle

From equation (3), we have

$$k = \frac{n\pi}{L} \quad \text{where } n = 1,2,3, \text{ etc.} \rightarrow (6)$$

$$\text{But } k^2 = \frac{2m}{\hbar^2} E$$

Squaring equation (6) and substituting for k^2

$$k^2 = \frac{n^2\pi^2}{L^2}$$

$$\text{i.e., } \frac{2m}{\hbar^2} E = \frac{n^2\pi^2}{L^2}$$

$$\text{or } E = \frac{n^2\pi^2 \hbar^2}{2mL^2} \quad \text{where } n = 1,2,3, \text{ etc.} \rightarrow (7)$$

For each value of n, there is an energy level E_n . Each value of E_n is called an Eigen value.

The above equation shows that a particle confined to a box cannot have any arbitrary or continuous value of energy. It can have only certain discrete values of energy specified by the equation, $E_n = \frac{n^2\pi^2 \hbar^2}{2mL^2}$. This means that the energy levels of a particle are quantized. The lowest possible energy level of the particle corresponding to $n=1$, is called the ground state of the particle. Another significant fact is that the particle cannot have zero energy.

HYDROGEN ATOM WAVE FUNCTIONS

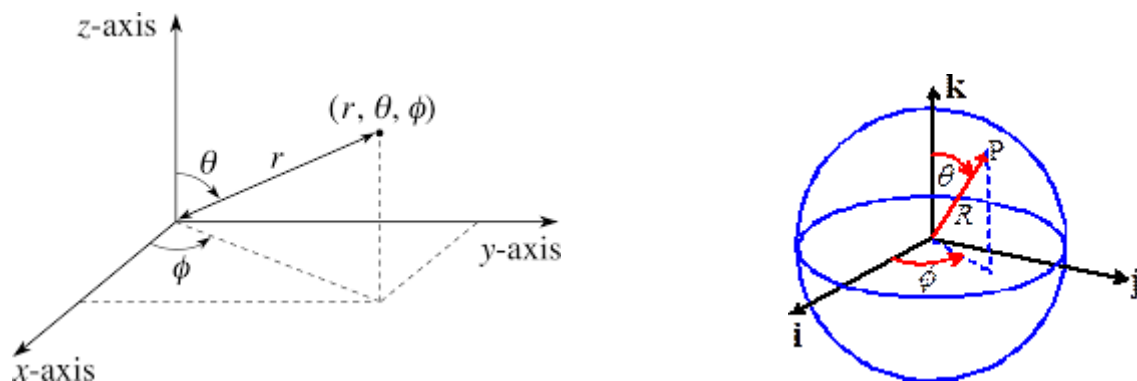
The hydrogen atom is the simplest atom consisting of one electron and one proton. The time-independent form of Schrodinger equation is $H\Psi = E\Psi$. The Hamiltonian operator consists of two terms, i.e., the kinetic energy term and the potential energy terms. The potential energy, V , of the coulombic interaction between the nucleus of charge $+e$ and the electron of charge $-e$ is

$$V = Q_1 Q_2 / 4\pi\epsilon_0 r, \text{ with } Q_1 = Ze = e \text{ (For hydrogen, } Z = 1) \text{ and } Q_2 = -e$$

i.e., $V = -\frac{e^2}{4\pi\epsilon_0 r}$ where $\epsilon_0 = 8.854 \times 10^{-12} \text{ J}^{-1}\text{C}^2\text{m}^{-1}$ is the vacuum permittivity. The negative sign indicates that the potential energy falls (becomes more negative) as the distance between the nucleus and the electron decreases. The Schrodinger equation for the hydrogen atom can then be written as

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} \left(E + \frac{e^2}{4\pi\epsilon_0 r} \right) \Psi = 0$$

This equation can be solved by transforming the Cartesian coordinates into spherical coordinates r , θ and ϕ (PHI), as shown in the figure



The wave function $\Psi(x,y,z)$ can be expressed as the product of two functions, one of which (the angular part Y or ψ) depends only on the angles θ and Φ the other of which (the radial part R) depends only on the distance from the nucleus. Thus, we have

$$\Psi(r, \theta, \Phi) = R(r) Y(\theta, \phi)$$

The Schrodinger equation after transformation into spherical coordinates takes the form

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2} \sin^2 \theta \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2} \sin^2 \theta \frac{\partial^2 \psi}{\partial \phi^2} + \frac{8\pi^2 m}{h^2} \left(E + \frac{e^2}{4\pi\epsilon_0 r} \right) \psi = 0$$

On solving this equation, we get three ordinary differential equations. Each of these differential equations on solving leads to the emergence of three quantum numbers.

1. The principal quantum number, n : This is a number that can assume any positive integral value, excluding zero. Its value determines the energy of the hydrogen atom by the formula

$$E = -\frac{me^4 Z^2}{8\epsilon_0^2 n^2 \hbar^2}$$

Where m and e are the mass and charge of electron, Z atomic number, $\hbar$ reduced Planck's constant ($\frac{h}{2\pi}$).

2. The orbital angular momentum quantum number or Azimuthal quantum number, l

The value of l determines the angular momentum of the electron. It may assume all integral values from 0 to $n-1$. i.e., 0, 1, ..., $n-2$, $n-1$.

3. The orbital magnetic quantum number, m.

The value of m indicates the magnetic field associated with the circulating electron. m can assume all integral values between $-l$ and $+l$, including zero. That is m can have $2l+1$ values of

$$-l, -l+1, \dots, 0, 1, \dots, l.$$

4. The electron has a spin angular momentum due to its spinning about its own axis. The magnetic quantum number associated with this spin has two possible values, $+\frac{1}{2}$ and $-\frac{1}{2}$.

Since the value of n restricts the possible values of l , and the value of l in turn restricts the allowed values of m , only certain combinations of quantum numbers are possible. In general, the number of possible combinations of quantum numbers all with the same value of n is $2n^2$. Each set of quantum numbers is associated with a different type of electronic motion. Each possible combination of quantum numbers represents various possible types of orbital. An orbital is the possibility of finding an electron at various points in an atom.

The orbitals are classified according to the values n and l associated with them. To avoid confusion over use of two numbers, the numerical values of l are replaced by letters:

electrons in orbitals with $l=0$ are called s electrons, those electrons occupying orbitals for which $l=1$ are p electrons, and those for which $l=2$ are called d electrons. The numerical and alphabetical correspondences are summarized in the table below:

n	l	m	Number of Orbitals	Orbital Name	Number of Electrons	Number of Electrons Per Orbital
1	0	0	1	1s	2	2
2	0	0	1	2s	2	2
	1	-1, 0, +1	3	2p	6	2
3	0	0	1	3s	2	2
	1	-1, 0, +1	3	3p	6	2
	2	-2, -1, 0, +1, +2	5	3d	10	2
4	0	0	1	4s	2	2
	1	-1, 0, +1	3	4p	6	2
	2	-2, -1, 0, +1, +2	5	4d	10	2
	3	-3, -2, -1, 0, +1, +2, +3	7	4f	14	2

Using the alphabetical notation for l , we can say that in the ground state of the hydrogen atom ($n=1, l=0$) we have 1s electron or that the electron moves in a 1s orbital.

Forms of hydrogen atom wave functions

The wave function ψ for the electron in hydrogen atom can be expressed as the product of two functions as

$$\Psi(r, \theta, \Phi) = R(r) Y(\theta, \phi)$$

The table shown below contains the expressions for the angular and radial parts of the one-electron –atom wave functions. Note that the angular part of the wave function for an s orbital is always the same, $(1/4\pi)^{1/2}$, regardless of principal quantum number. Thus, all orbitals of a given type (s, p, or d) have the same angular behavior. The radial part of the wave function, however, depends both on the principal quantum number n and on the angular momentum quantum number l .

n	ℓ	m_ℓ		$F(\phi)$	$P(\theta)$	$R(r)$	☒ $n=1,2$ ☐ $n=3$
1	0	0	1s	$\frac{1}{\sqrt{2\pi}}$	$\frac{1}{\sqrt{2}}$	$\frac{2}{a_0^{3/2}} e^{-r/a_0}$	☒ Separated
2	0	0	2s	$\frac{1}{\sqrt{2\pi}}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2\sqrt{2} a_0^{3/2}} \left[2 - \frac{r}{a_0} \right] e^{-r/2a_0}$	☐ Combined
2	1	0	2p	$\frac{1}{\sqrt{2\pi}}$	$\frac{\sqrt{6}}{2} \cos \theta$	$\frac{1}{2\sqrt{6} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$	
2	1	± 1	2p	$\frac{1}{\sqrt{2\pi}} e^{i\phi}$	$\frac{\sqrt{3}}{2} \sin \theta$	$\frac{1}{2\sqrt{6} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$	

$$a_0 = \frac{\hbar^2}{me^2} = 0.0529 \text{ nm} = \text{first Bohr radius}$$

To find the wave function for a particular orbital, we simply multiply the appropriate angular and radial parts together.

Wave function for 1s orbital

For the ground state, $n=1$, and $l=0$, $m=0$, Such an orbital is called 1s orbital. It depends only on the radial coordinate (r) and is independent of θ and ϕ . The normal wave function for 1s orbital in the ground state is given below:

$$\psi = \frac{1}{\sqrt{\pi}} (1/a_0)^{3/2} e^{-r/a_0}$$

$$\text{where } a_0 = \frac{\hbar^2(4\pi\epsilon_0)}{4\pi^2 m e^2} = 0.529 \times 10^{-8} \text{ cm (Bohr radius)}$$

By squaring this function, we obtain an expression for the probability of finding the electron in a unit volume as a function of r , the distance from the nucleus. Thus, the probability of finding an electron in a 1s orbital is independent of the angular coordinates θ and ϕ .

Wave function for 2s orbital

When $n=2$ and $l=0$, the orbital is the 2s orbital and it has the wave function

$$\psi(2s) = \frac{1}{4\sqrt{2\pi}} \left(\frac{1}{a_0} \right)^{3/2} \left(2 - \frac{r}{a_0} \right) e^{-\frac{r}{2a_0}}$$

$\frac{r}{2a_0}$ in the exponential factor shows that as r increases, the 2s function decreases in amplitude more slowly than does the 1s function, which has $\frac{r}{a_0}$ in its exponential factor.

When $r = 2a_0$, the factor $(2 - \frac{r}{a_0})$ becomes zero and hence the wave function ψ is zero. That is

$r = 2a_0$ is the condition for a **radial node**, i.e., the radius at which the probability of finding an electron is zero.

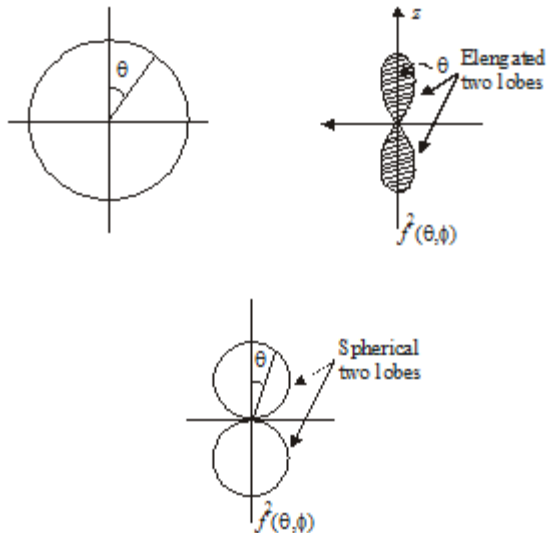
Similarly, on analyzing the wave function for 3s orbital it can be shown that there are two radial nodes for 3s orbital. In general, for an ns orbital, there are $n-1$ radial nodes.

Wave function for 2p orbital

When $n=2$ and $l=1$, we have p orbitals. The wave function for p orbital is dependent on θ and ϕ and hence p orbitals have marked directional character. For states $l=1$, m can be $-1, 0, +1$. Different values of m correspond to orbitals of different orientation of orbital angular momentum. The normalized wave function for the p_z orbital (the orbital having zero component of angular momentum about z axis) can be expressed as

$$\psi(2p_z) = \frac{1}{4\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \frac{r}{a_0} e^{-\frac{r}{2a_0}} \cos \theta$$

In contrast to the s orbital wave functions, which were nonzero at $r=0$, the p orbital wave functions vanish at $r=0$. Thus, it can be said that s electron has the ability to penetrate to the nucleus while the p electrons get close to the nucleus but does not penetrate to it. Again, in contrast to s orbitals, the p orbitals are not spherically symmetric, but have directional character due to the angular part of the 2p wave function. Since $\psi(2p_z)$ is proportional to $\cos \theta$, it has angular maximum along the positive z axis, for there $\theta=0$, and $\cos 0 = 1$, which is the maximum value of cosine. Similarly, along the negative z axis the $2p_z$ function has its most negative value, for there, $\theta=\pi$, and $\cos \pi = -1$, the most negative value of the cosine.



Radial probability distribution

The amplitude or intensity of three-dimensional electron wave, known as Wavefunction is represented by $\psi_{nlm}(r, \theta, \Phi)$. It has both radial and angular parts.

$$\psi_{nlm}(r, \theta, \Phi) = \text{Radial part} \times \text{Angular part} = R_{nl}(r) \times \Phi_{lm}(\theta, \Phi)$$

where,

$R_{nl}(r)$ = Radial wavefunction

$\Phi_{lm}(\theta, \Phi)$ = Angular wavefunction

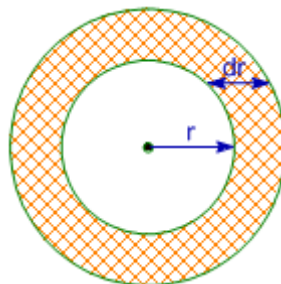
Radial probability density: The square of the radial wavefunction is known as radial probability density.

$$\text{Radial probability density} = |R|^2$$

It gives the probability of finding the electron at one particular point a distance r from the nucleus

Radial probability distribution or Radial probability function:

It is the probability of finding the electron anywhere in a spherical shell enclosed between a sphere of radius ' $r + dr$ ' and a sphere of radius ' r ' from the nucleus.

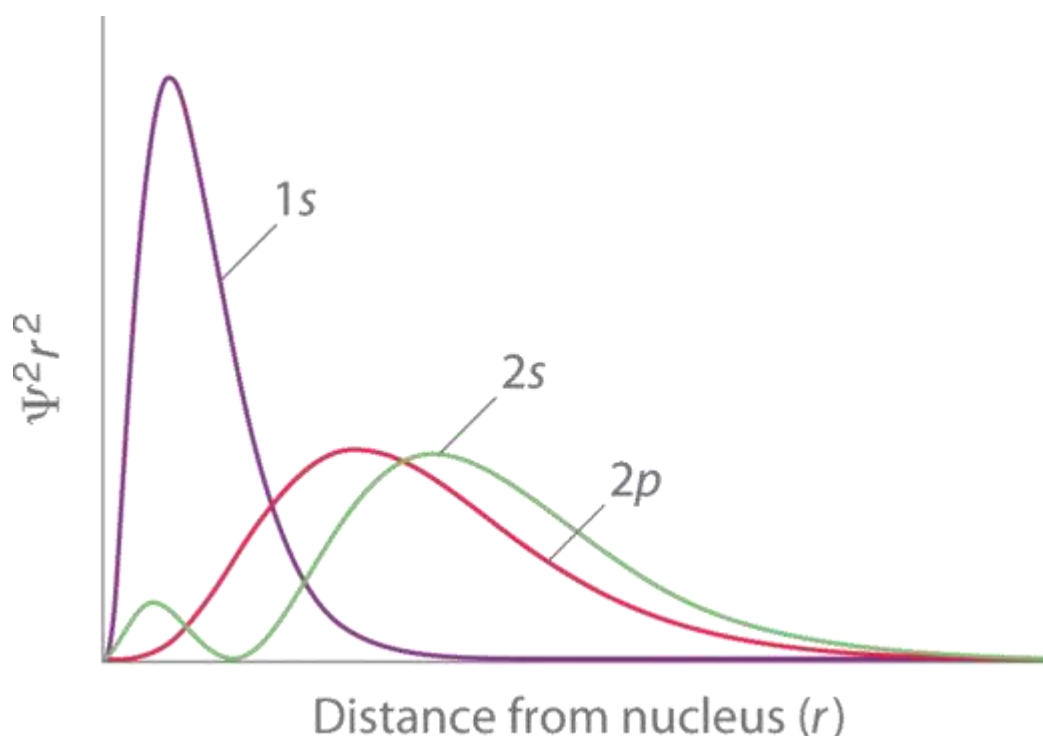


The relation between radial probability and radial probability density is given as:

$$\begin{aligned}\text{Radial Probability} &= \text{Radial Probability Density} \times \text{Volume of spherical shell} = 4\pi r^2 dr R^2 \\ &= 4\pi r^2 R^2 \text{ when } dr = 1\end{aligned}$$

Plots of hydrogen atom wave functions OR Radial probability distribution curve

Radial probability distribution curve gives idea about the distribution of electron density at a radial distance around the nucleus without considering the direction or angle. The plots given below show how the radial probability ($4\pi r^2 R^2$ or $\psi^2 r^2$) depends on the distance from the nucleus for various orbitals.



The maxima in the radial probability curves occur at radii at which the electron is most likely to be found. For the 1s electron of the hydrogen atom, this radius of maximum probability is equal to a_0 (Bohr radius = 0.529×10^{-8} cm). The curves show that on average the 2s electrons spends its time a greater distance from the nucleus than does a 1s electron. This is consistent with the relative energies of the 1s and 2s states, because the electron that is closer to the nucleus is bound to have the lower energy. Comparison of the radial probability curves of electrons with the same value of n but different values of l shows that the average distance of such electrons from the nucleus is approximately the same. However, an s electron has a greater chance of being very close to the nucleus than does a p electron.

Molecular orbital theory

Molecular orbital theory (MO Theory) is a comprehensive approach to chemical bonding based on quantum mechanical principles. It considers the entire molecule as one unit – a collection of nuclei and electrons. All the electrons are moving under the influence of all the nuclei and all the electrons. The basic idea of MO Theory may be summarized as follows:

1. Each electron in a molecule is described by a wave function known as molecular orbital (MO). These molecular orbitals will be associated with all the nuclei present in the molecule. So molecular orbitals are described as polycentric orbitals.
2. When two atomic orbitals combine two molecular orbitals are formed, namely bonding molecular orbital and antibonding molecular orbital.
3. The bonding molecular orbital possess lower energy with greater stability than the corresponding antibonding molecular orbital. Electrons fill up the molecular orbitals following Pauli's Aufbau, and Hund's rule just like atomic orbitals.

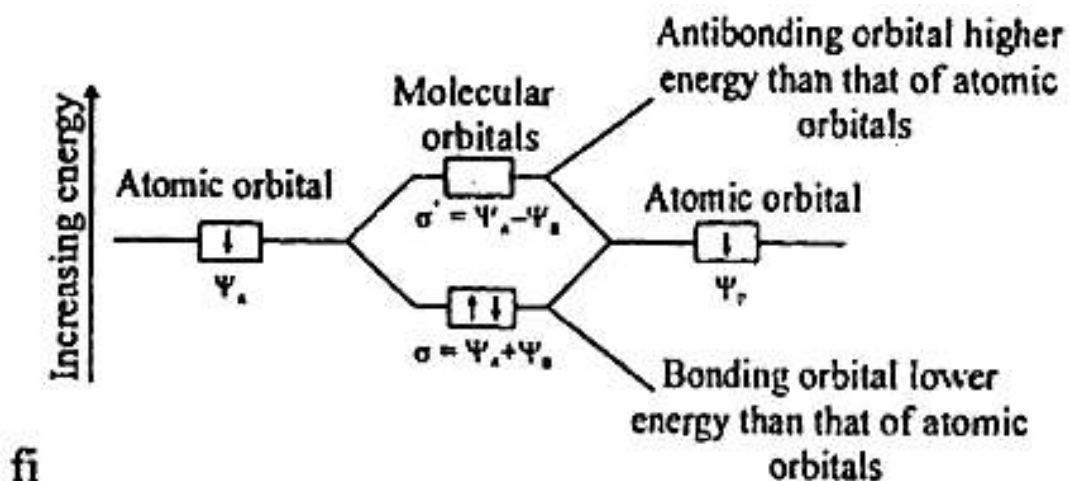
LCAO method

One of the most important ways for constructing molecular orbitals is an approximation scheme known as Linear combination of atomic orbitals (LCAO). As per LCAO, molecular orbitals are formed by combination of the atomic orbitals of the combining atoms. The number of molecular orbitals formed must always be equal to the number of atomic orbitals that are combined. The electronic wave functions, namely ψ_1 and ψ_2 for the combining atoms can interact in a constructive or destructive manner to form molecular orbitals. If there is a constructive (or in phase) combination of atomic orbitals, an increase in electron probability occurs between the nuclei of the approaching atoms leading to the formation of energetically favorable bonding molecular orbitals denoted as ψ_B . In destructive (or out of phase) combination of atomic orbitals, there is a zero-electron probability between the nuclei of approaching atoms leading to the formation of energetically unfavorable antibonding molecular orbitals denoted as ψ_A . The bonding molecular orbitals are formed by adding wave functions of electrons in the two atomic orbitals, whereas antibonding molecular orbitals are formed by subtracting their wave functions as

$$\psi_B = \psi_1 + \psi_2 \quad \text{and}$$

$$\psi_A = \psi_1 - \psi_2$$

The relative energy levels of atomic and molecular orbitals are illustrated in the following diagram.



σ and π molecular orbitals

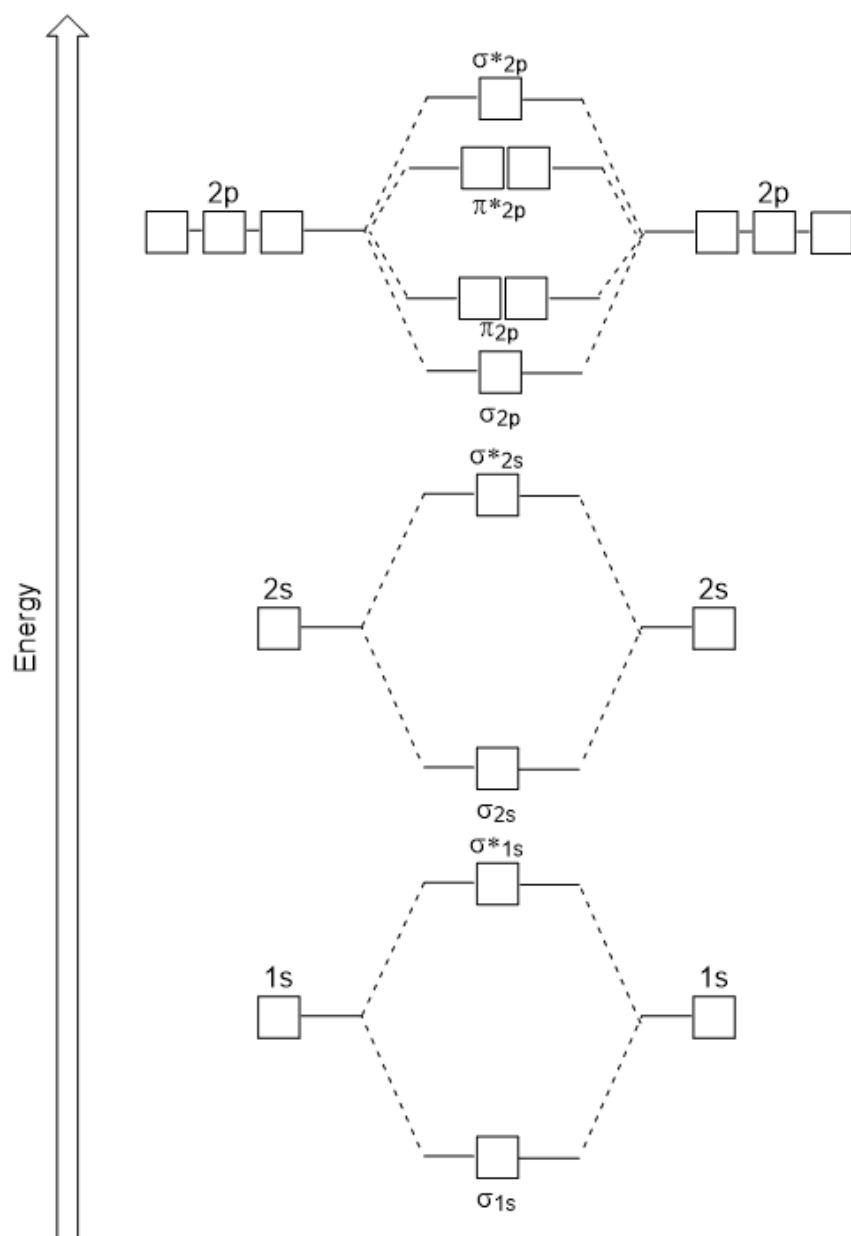
Molecular orbitals can be sigma or pi depending on the mode of overlap of atomic orbitals. If a linear or head-on collision occurs between two atomic orbitals, sigma (σ) molecular orbitals will be formed. Lateral or side - to - side overlapping of p orbitals produce pi (π) molecular orbitals.

Two or more molecular orbitals of same energy are called degenerate molecular orbitals.

E.g., π 2p orbitals are doubly degenerate as there are two orbitals of equal energy $\pi 2p_y = \pi 2p_z$ and their antibonding molecular orbitals are also degenerate; $\pi^* 2p_y = \pi^* 2p_z$.

The sequence of energy levels in the increasing order of energy that helps in predicting the electronic structure of simple molecule is as follows:

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_x < \pi 2p_y = \pi 2p_z < \pi^* 2p_y = \pi^* 2p_z < \sigma^* 2p_x$$



Bond order

The difference between the number of bonding and antibonding electrons that is divided by 2 is called bond order.

$$\text{Bond order} = \frac{\text{Number of bonding electrons} - \text{Number of antibonding electrons}}{2}$$

Thus, for a simple molecule such as hydrogen that has two electrons, its bond order will be

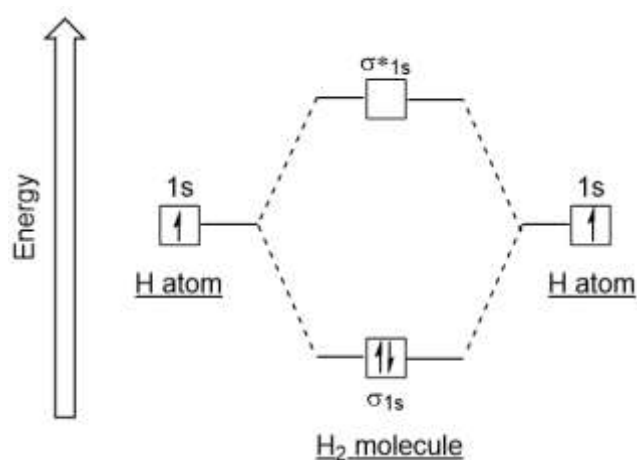
$\frac{2-0}{2} = 1$. This indicates that H₂ molecule has one bond. However, it is not necessary that the bond order will always be a whole number.

Energy level diagrams of diatomic molecules

A) Homonuclear diatomic molecules

H₂ molecule

The simplest homonuclear diatomic molecule is formed when the 1s atomic orbitals of two hydrogen atoms combine. The electrons occupy the molecular orbital of the lowest energy σ 1s bonding orbital. A molecular orbital can hold two electrons, so both electrons in the H₂ molecule are in the σ 1s bonding orbital and the electronic configuration is $(\sigma 1s)^2$

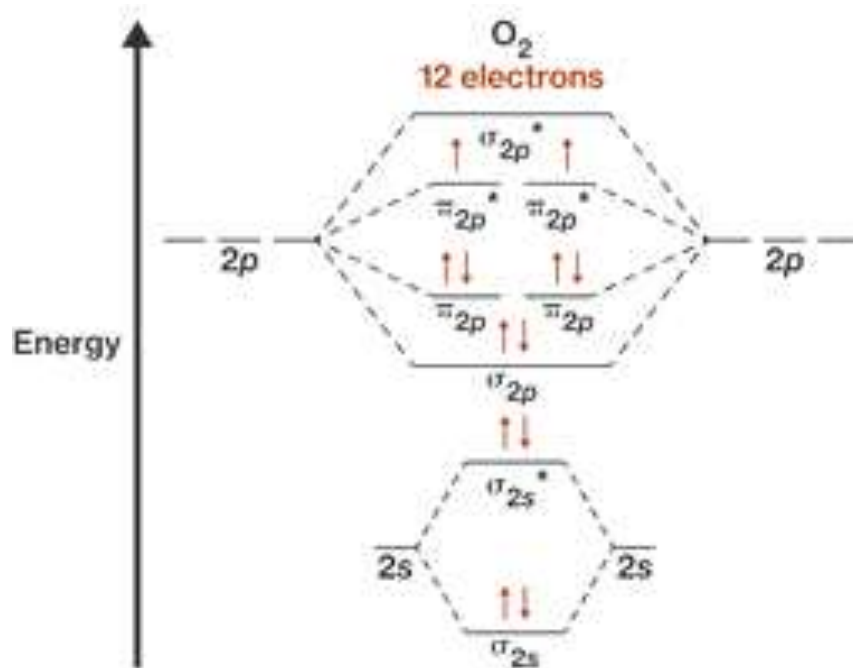


$$\text{Bond order for H}_2\text{molecule} = \frac{1}{2} (N_b - N_a) = \frac{2-0}{2} = 1$$

O₂ molecule

The ground state electronic configuration of oxygen atom is $1s^2, 2s^2, 2p^4$, and electronic configuration of O₂ molecule is $[\text{KK}, (\sigma 2s)^2, (\sigma^* 2s)^2, (\sigma 2p_x)^2, (\pi 2p_y = \pi 2p_z)^4, (\pi^* 2p_y = \pi^* 2p_z)^2]$

The 1s electrons from both the oxygen atoms are referred to as K shell electrons since they do not participate in bonding as they are in the inner shell and denoted as KK in the electronic configuration.

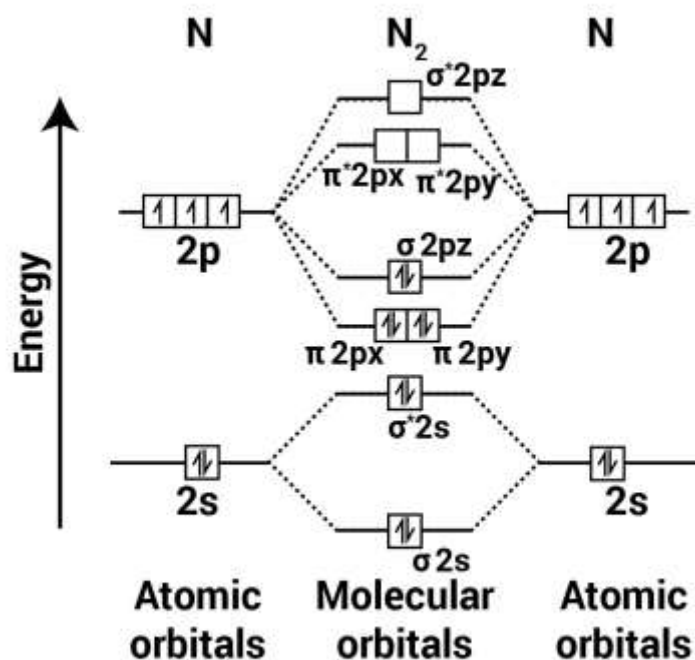


The bond order is given as: $\frac{1}{2} (N_b - N_a) = \frac{8-4}{2} = 2$

Thus the oxygen molecule has a double bond with two unpaired electrons, and thus it exhibits paramagnetism.

N₂ molecule

The ground state electronic configuration of nitrogen atom is $1s^2, 2s^2, 2p^3$ and the electronic configuration of nitrogen molecule is $[KK, (\sigma 2s)^2, (\sigma^* 2s)^2, (\pi 2p_x = \pi 2p_y)^4, (\sigma 2p_z)^2]$



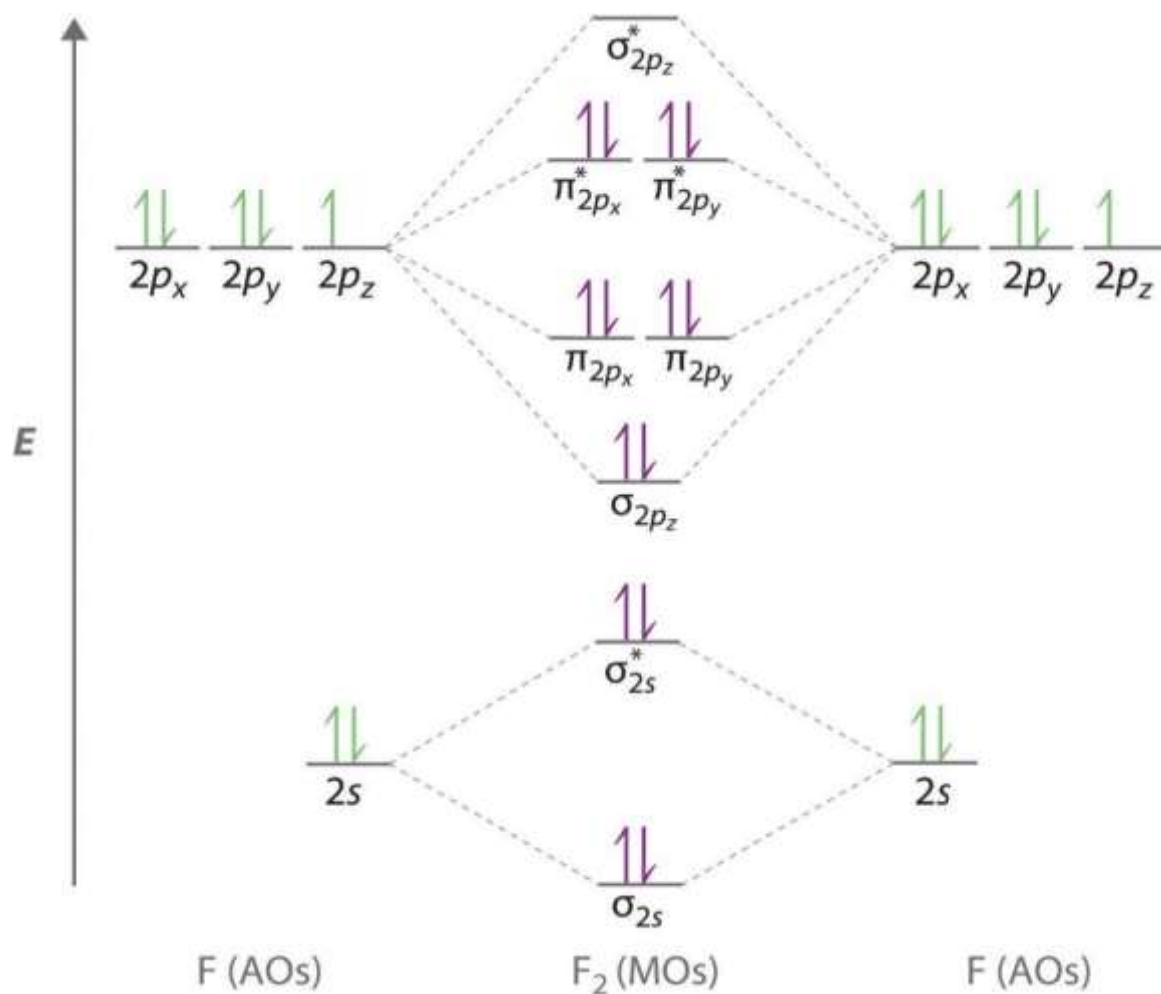
For the N_2 molecule the molecular orbitals fall in a slightly different pattern, as shown in the figure above. Here, the σ_{2p} orbital lies a bit higher in energy than the two bonding π_{2p} orbitals. This feature is a consequence of the repulsion between electrons of σ_{2s} and σ_{2p} orbital. These repulsions, and the resultant elevation of the energy of the σ_{2p} orbital, occur because both the σ_{2p} and σ_{2s} electrons tend to occupy the same region of space in this light diatomic molecule. This effect is diminished in O_2 and F_2 because in the atoms O and F, the $2s$ atomic orbitals are quite low in energy and are largely confined to regions close to the nucleus.

The bond order (i.e., the number of covalent bonds) is given as: $\frac{1}{2} (N_b - N_a) = \frac{8-2}{2} = 3$

Hence nitrogen is a triple bond molecule ($N \equiv N$) with one sigma and two pi bonds with diamagnetic properties.

F_2 molecule

The ground state electronic configuration of fluorine atom is $1s^2, 2s^2, 2p^5$, and electronic configuration of F_2 molecule is $[KK, (\sigma_{2s})^2, (\sigma_{2s}^*)^2, (\sigma_{2p_x})^2, (\pi_{2p_y} = \pi_{2p_z})^4, (\pi_{2p_y}^* = \pi_{2p_z}^*)^2]$.

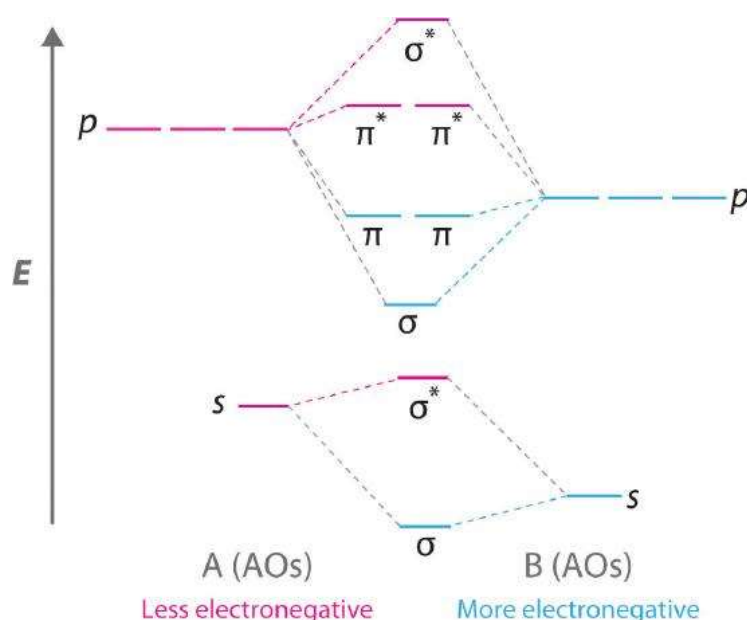


The bond order is given as: $\frac{1}{2} (N_b - N_a) = \frac{8-6}{2} = 1$

Thus, the F_2 molecule has a single bond with no unpaired electrons, and thus it exhibits diamagnetism.

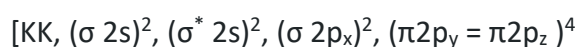
B) Heteronuclear diatomic molecules

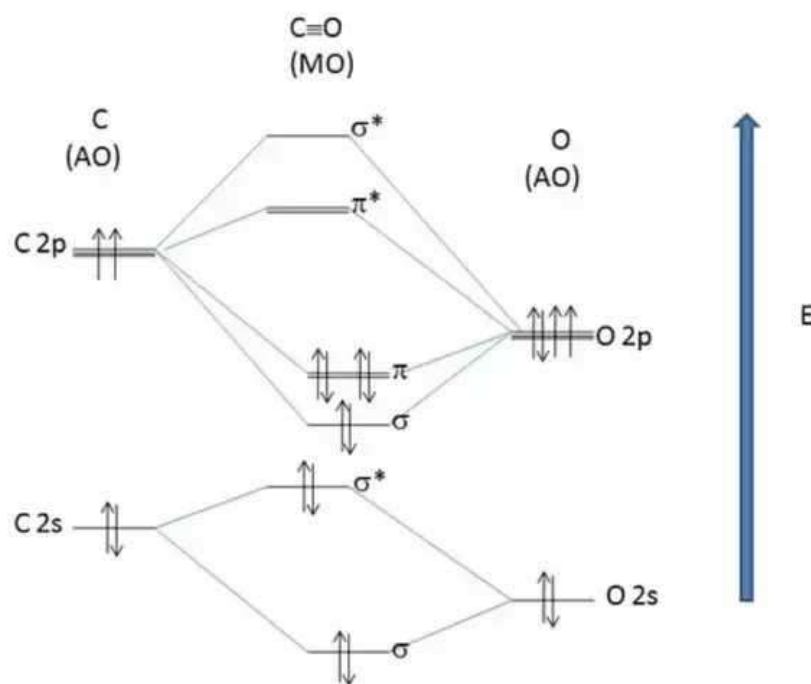
If two bonded atoms in a molecule are of different elements, they are called heteronuclear diatomic molecules. E.g., CO, HCl, NO, etc. The two atomic orbitals of the combining elements are at different energies due to differing electronegativities between atoms. If the electronegativity of atom B is greater than the electronegativity of atom A, then the energy of atomic orbitals of A will be greater than the energy of atomic orbitals of B. Then the atomic orbital of B will be making greater contribution to the bonding molecular orbital because it is having a lower energy and it lies close to the bonding molecular orbital. Similarly, the atomic orbitals of A will be making greater contribution to the antibonding molecular orbital because it has a higher energy and it lies close to the antibonding molecular orbital.



Carbon monoxide molecule

The electronic configuration of C and O atoms are ${}_6\text{C} = 1s^2, 2s^2, 2p^2$ and ${}_8\text{O} = 1s^2, 2s^2, 2p^4$. The number of electrons available for bonding from carbon and oxygen are 4 and 6 respectively; thereby ten electrons need to be accommodated in the molecular energy levels. The electronic configuration of carbon monoxide molecule can be expressed as





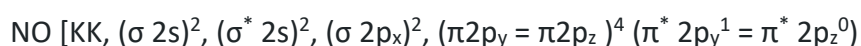
As shown in the figure, the bonding resulting from $(\sigma 2s)^2$ is effectively cancelled by antibonding $(\sigma^* 2s)^2$. As all the six electrons are present in the bonding molecular orbitals and none of it is in the antibonding molecular orbitals, the bond order in CO molecule is

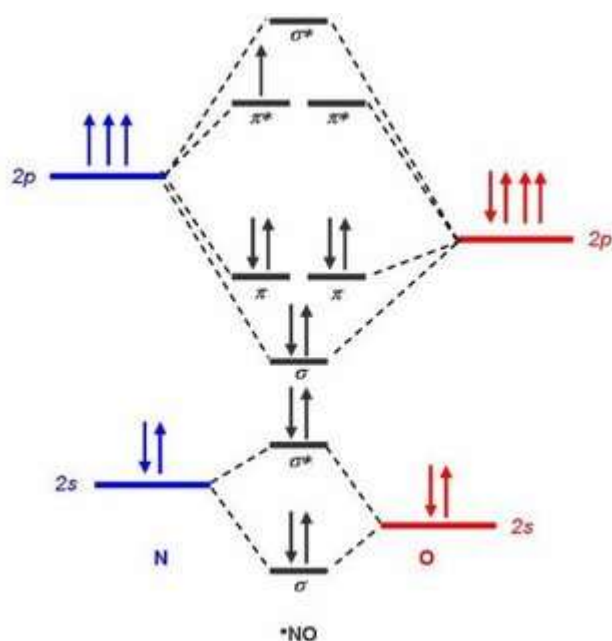
$$\frac{1}{2} (N_b - N_a) = \frac{8-2}{2} = 3$$

Thus in CO molecule, there is a triple bond $C \equiv O$ with one sigma and two pi bonds. All the electrons are paired and hence CO is a diamagnetic molecule.

Nitric oxide molecule

The electronic configuration of N and O atoms are ${}_7N = 1s^2, 2s^2, 2p^3$ and $O_8 = 1s^2, 2s^2, 2p^4$. Thus, in NO there are 11 electrons to be filled in molecular orbitals. Thus, the configuration of NO molecule can be written as





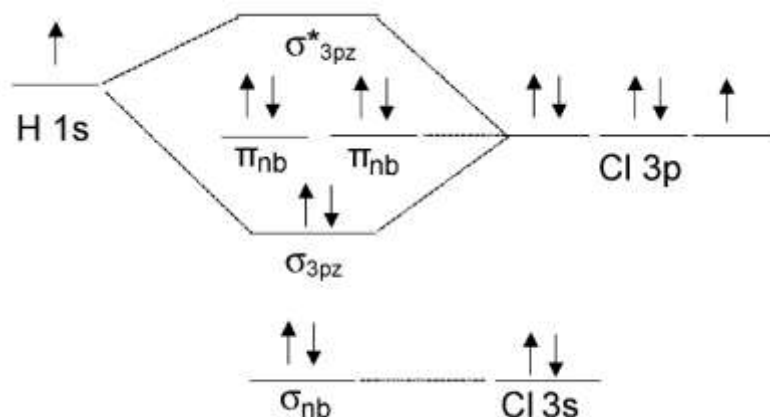
The four electrons of the 2s orbitals fill up σ 2s and σ^* 2s molecular orbitals. Out of the remaining seven electrons, six electrons fill up the bonding molecular orbitals (i.e., σ 2p_x, π 2p_y and π 2p_z). The remaining single electron will occupy the antibonding molecular orbital π^* 2p_y. Due to the presence of single electron, NO molecule exhibits paramagnetism.

the bond order in NO molecule is

$$\frac{1}{2} (N_b - N_a) = \frac{6-1}{2} = 2.5$$

Hydrogen chloride molecule

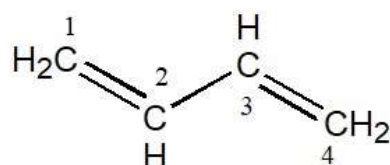
The electronic configuration of H and Cl atoms are: ${}_1\text{H} = 1s^1$ and ${}_{17}\text{Cl} = 1s^2, 2s^2, 2p^6, 3s^2, 3p^2_x, 3p^2_y, 3p^1_z$. Because the energies of the 3s, 3p_x, and 3p_y orbitals of chlorine do not change when HCl forms, they are called **nonbonding molecular orbitals**. A nonbonding molecular orbital occupied by a pair of electrons is the molecular orbital equivalent of a lone pair of electrons.



By definition, electrons in nonbonding orbitals have no effect on bond order, so they are not counted in the calculation of bond order. Thus, the predicted bond order of HCl is $(2 - 0) \div 2 = 1$. Because the σ bonding molecular orbital is closer in energy to the Cl $3p_z$ than to the H $1s$ atomic orbital, the electrons in the σ orbital are concentrated closer to the chlorine atom than to hydrogen.

Pi – molecular orbitals of butadiene

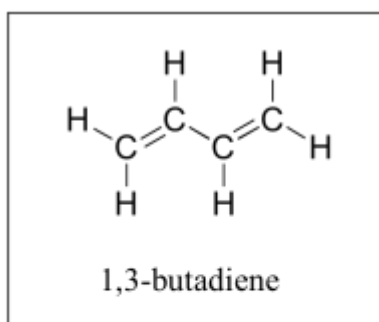
1,3-butadiene is the simplest example of a system of **conjugated pi bonds**. To be considered conjugated, two or more pi bonds must be separated by only one single bond



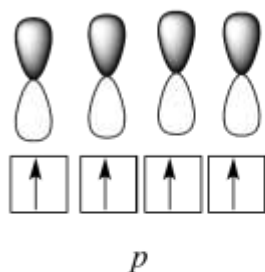
Experimentally, however, it is observed that there is a significant barrier to rotation about the C_2 - C_3 sigma bond. In addition, the C_2 - C_3 bond is 148 pm long, shorter than a typical carbon-carbon single bond (about 154 pm), though longer than a typical double bond (about 134 pm).

Molecular orbital theory accounts for these observations with the concept of **delocalized pi bonds**. 1, 3-Butadiene is having sp^2 hybridized carbon atoms. The three hybridized orbitals will form sigma bonds with two H atom and one C atom. The remaining unpaired unhybridized electron of p orbital is involving in the π bond formation. Thus, four carbon atoms are having four unpaired unhybridized p electron.

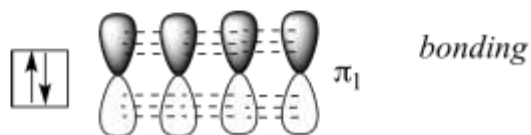
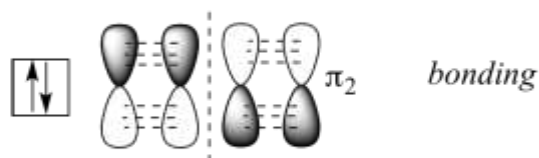
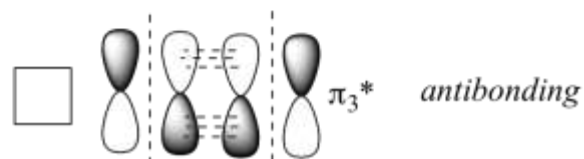
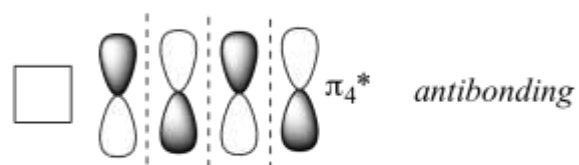
The four $2p$ atomic orbitals combine mathematically to form four pi molecular orbitals of increasing energy. Two of these – the bonding pi orbitals – are lower in energy than the p atomic orbitals from which they are formed, while two – the antibonding π^* orbitals – are higher in energy.



atomic orbitals



molecular orbitals

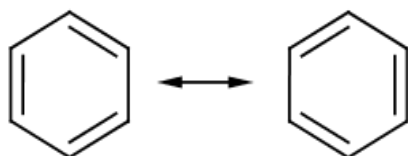


The lowest energy molecular orbital, π_1 , has only constructive interaction and zero nodes. Higher in energy, but still lower than the isolated p orbitals, the π_2 orbital has one node but two constructive interactions – thus it is still a bonding orbital overall. Looking at the two antibonding orbitals, π_3^* has two nodes and one constructive interaction, while π_4^* has three nodes and zero constructive interactions.

By the *aufbau* principle, the four electrons from the isolated $2p$ atomic orbitals are placed in the bonding π_1 and π_2 MO's. Because π_1 includes constructive interaction between C_2 and C_3 , there is a degree, in the 1,3-butadiene molecule, of pi-bonding interaction between these two carbons, which accounts for its shorter length and the barrier to rotation. The valence bond picture of 1,3-butadiene shows the two pi bonds as being isolated from one another, with each pair of pi electrons 'stuck' in its own pi bond. However, molecular orbital theory predicts (accurately) that the four pi electrons are to some extent delocalized, or 'spread out', over the whole pi system

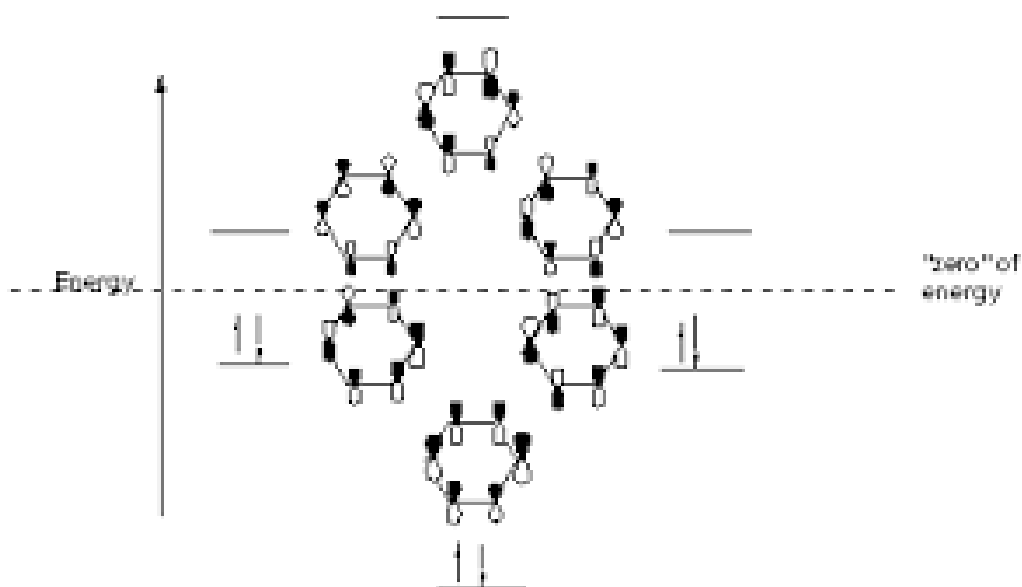
Pi – molecular orbitals of benzene

Although benzene is most often represented with three double bonds and three single bonds, all the carbon-carbon bonds are equal in length (1.39 Å) which is in between a single (1.54 Å) and double bond (1.34 Å).



Benzene is unusually stable and, in spite of the presence of three double bonds it does not give the typical addition reactions of alkenes. The Molecular orbital theory is especially helpful in explaining these unique properties of benzene and other **aromatic** compounds

Benzene, C_6H_6 , has a planar hexagonal structure in which all the carbon atoms are sp^2 hybridized. There is an unhybridized 2p orbital on each carbon atom perpendicular to the ring which is used to form the molecular orbitals. Since benzene has six contributing p-orbitals, we should expect six pi orbitals for each.



The lowest energy level (π_1) has all the p-orbitals aligned the same way, and therefore has zero nodes between the p orbitals. (A “node” is where there is a change in phase between adjacent p-orbitals). It is strongly bonding because the constructive interference between

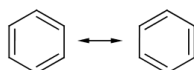
neighboring p orbitals (in phase combination) results in good accumulation of electron density between the nuclei. The highest energy molecular orbital (π_6) results from destructive interference between neighbors (out of phase combination) and the molecular orbital has a nodal plane between each pair of neighbors. Among the four intermediate molecular orbitals, the $\pi_2 - \pi_3$ and $\pi_4^* - \pi_5^*$ pairs are called **degenerate**, meaning they are at the same energy level. The π_2 and π_3 molecular orbitals have nodal planes between two neighbouring pairs and are net bonding while π_4^* and π_5^* molecular orbitals have nodal planes between four neighbouring pairs and are net antibonding.

There are six electrons to be accommodated (one is supplied by each C atom) and they occupy the lowest three orbitals to give the configuration $\pi_1^2, (\pi_2 = \pi_3)^4$. It is an important feature of the configuration that the only molecular orbitals occupied have a net bonding character. This is one contribution to the stability or low energy of benzene molecule. Another feature of the molecular orbital description of benzene is that each molecular orbital spreads around the C_6 ring. That is, π bonding is delocalized, and each electron pair helps to bind together all the C atoms.

Aromaticity

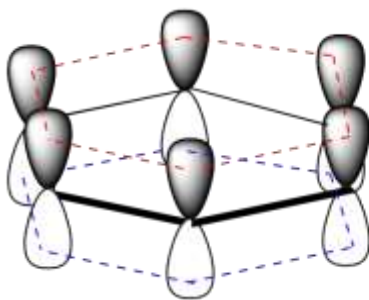
Aromatic compounds are essential in industry to produce important chemicals and polymers, such as polyester and nylon. Various biomolecules like proteins, DNA, RNA, etc also involve aromatic compounds as constituents.

Molecular orbital theory is especially helpful in explaining the unique properties of **aromatic** compounds such as benzene. Although benzene is most often drawn with three double bonds and three single bonds, in fact all of the carbon-carbon bonds are exactly the



Same. In addition, the pi bonds in benzene are significantly less reactive than 'normal' pi bonds. The peculiar structure of benzene and related compounds which is responsible for their special stability is referred to as 'aromaticity'.

The basis of the 'aromaticity' of benzene is that it is a *cyclic* molecule in which all of the ring atoms are sp^2 -hybridized. This allows the pi electrons to be delocalized in molecular orbitals that extend all the way around the ring, above and below the plane. For this to happen, of course, the ring must be planar – otherwise the p orbitals couldn't overlap properly. Benzene is indeed known to be a flat molecule.



π electrons delocalized around the ring, above and below the plane

The six π electrons in benzene occupy the lowest three bonding orbitals to give the configuration $\pi_1^2, (\pi_2 = \pi_3)^4$. Thus, in benzene, the bonding orbitals are completely filled, and the antibonding orbitals are empty. a full set of bonding MO's is similar in many ways to the 'full shell' of electrons in the atomic orbitals of the stable noble gases, helium, neon, and argon. **This is one contribution to aromaticity of benzene molecule.**

The general requirements for a compound to be aromatic are:

- 1.The compound must be cyclic
- 2.The ring must be planar.
- 3.Each atom in the ring must be sp^2 -hybridized.
- 4.Each element within the ring must have a p-orbital that is *perpendicular* to the ring.
- 5.The compound must follow **Hückel's Rule** which states that the ring has to contain $4n+2$ p-orbital electrons or π electrons where n is zero or a positive whole number